COMSATS University Islamabad
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ROLL NO:

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CLASS:

BSE-8B

SUBJECT:

DEVOPS FOR CLOUD COMPUTING

SUBMITTED TO:

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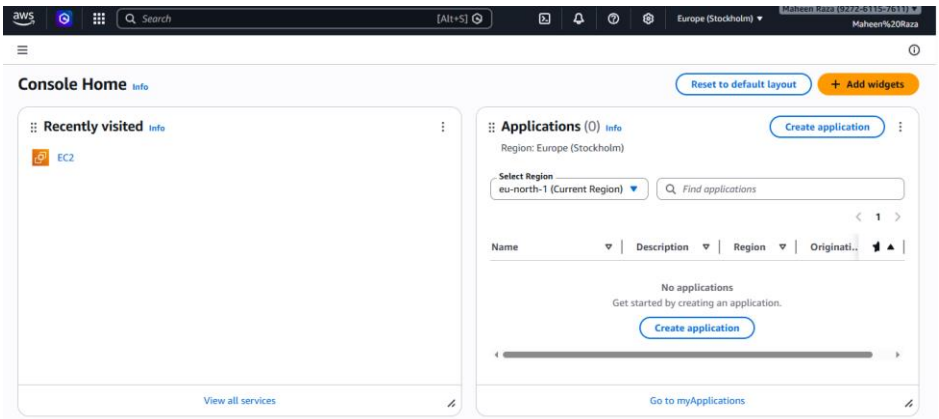
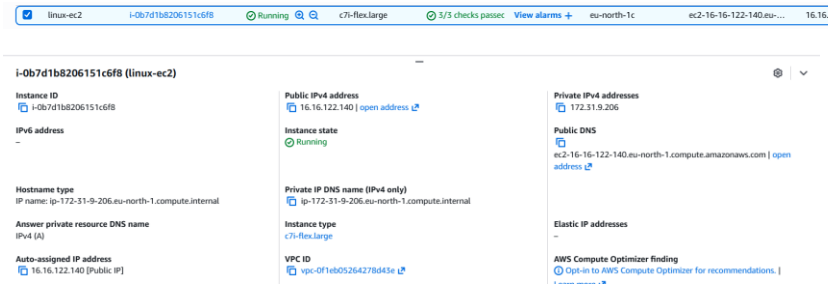
SUBMISSION DATE:

3/26/ 2026

Ansible Installation and 2 Automation tasks on single Linux machine



Step 1 - AWS EC2 Instance Create



Step 2 - Local Terminal (WSL / Ubuntu)

Project Folder Create

- mkdir ~/ansible-web
- cd ~/ansible-web

```
maheen@Maheen-Dell:~$ mkdir ~/ansible-web
maheen@Maheen-Dell:~$ cd ~/ansible-web
maheen@Maheen-Dell:~/ansible-web$
```

Files Create

- touch hosts.ini
- touch web_setup.yml

touch cv.html

```
maheen@Maheen-Dell:~$ mkdir ~/ansible-web
maheen@Maheen-Dell:~$ cd ~/ansible-web
maheen@Maheen-Dell:~/ansible-web$ touch hosts.ini
maheen@Maheen-Dell:~/ansible-web$ touch web_setup.yml
maheen@Maheen-Dell:~/ansible-web$ touch cv.html
maheen@Maheen-Dell:~/ansible-web$
```

```
maheen@Maheen-Dell:~/ansible-web$ ls
cv.html  hosts.ini  web_setup.yml
maheen@Maheen-Dell:~/ansible-web$
```

Nano hosts.ini file code:

```
GNU nano 7.2                                                    hosts.ini
[webservers]
16.16.122.140 ansible_user=ubuntu ansible_ssh_private_key_file=~/.ssh/aws-key.pem
```

Nano web_setup.yml file code:

```
GNU nano 7.2                                                    web_setup.yml
---
- name: Deploy CV Website and Create Users
  hosts: webservers
  become: yes

  tasks:

    - name: Install nginx
      apt:
        name: nginx
        state: present
        update_cache: yes

    - name: Start nginx
      service:
        name: nginx
        state: started
        enabled: yes

    - name: Copy CV HTML page
      copy:
        src: cv.html
        dest: /var/www/html/index.html

    - name: Create user1
      user:
        name: user1
        password: "{{ 'user123' | password_hash('sha512') }}"
        state: present
        create_home: yes

    - name: Create user2
      user:
        name: user2
        password: "{{ 'user123' | password_hash('sha512') }}"
        state: present
```

Step 3 - SSH Test

- ssh -i ~/.ssh/aws-key.pem ubuntu@16.16.122.140

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Mar 26 06:41:17 UTC 2026

  System load:  0.28              Temperature:           -273.1 C
  Usage of /:   88.9% of 6.71GB   Processes:             171
  Memory usage: 20%               Users logged in:       1
  Swap usage:   0%                IPv4 address for ens5: 172.31.9.206

  => / is using 88.9% of 6.71GB

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


*** System restart required ***
Last login: Thu Mar 26 06:16:26 2026 from 103.129.14.177
ubuntu@ip-172-31-9-206:~$
```

Step 4 - hosts.ini pem path

```
  GNU nano 7.2                                                hosts.ini
[webservers]
16.16.122.140 ansible_user=ubuntu ansible_ssh_private_key_file=~/.ssh/aws-key.pem
```

Step 5 - Connection test

- ansible webservers -i hosts.ini -m ping

```
16.16.122.140 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
iqra@DESKTOP-TEDRU1B:~/ansible-web$
```

Step 6 - Run Playbook

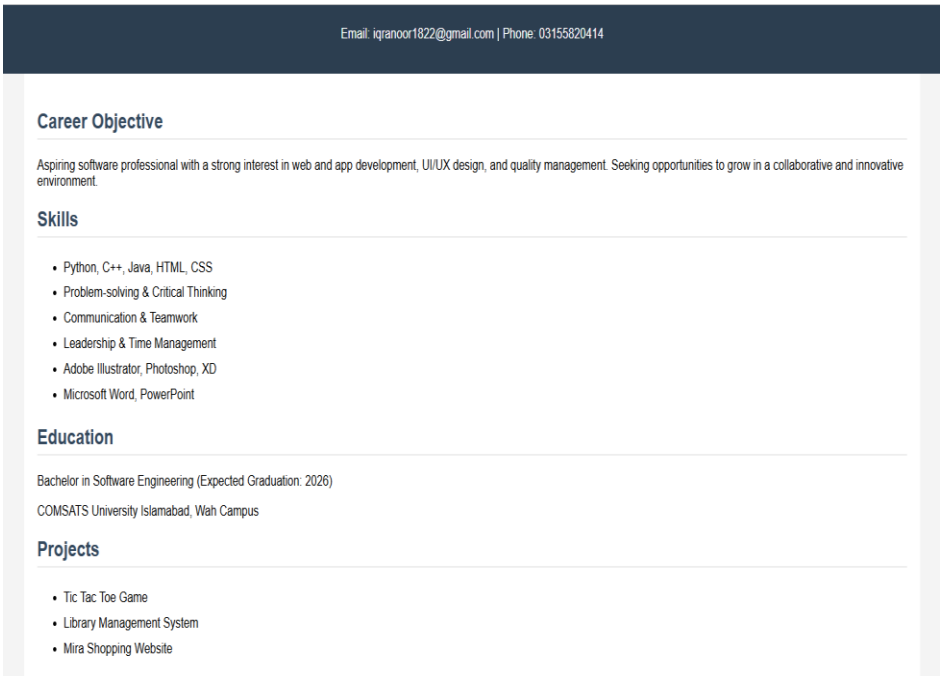
- ansible-playbook -i hosts.ini web_setup.yml

```
PLAY [Install Web Server and Create Users] *************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [16.16.122.140]

TASK [Install nginx] ***********************************************************************************************
changed: [16.16.122.140]

TASK [Start nginx] *************************************************************************************************
ok: [16.16.122.140]

TASK [Copy CV page] ************************************************************************************************
changed: [16.16.122.140]

TASK [Create user1] ************************************************************************************************
changed: [16.16.122.140]

TASK [Create user2] ************************************************************************************************
changed: [16.16.122.140]

PLAY RECAP *********************************************************************************************************
16.16.122.140              : ok=6    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Step 7 - Open Website

Browser:

- http://16.16.122.140



Certifications

- Student Week Certification
- Olympiad Math Certification
- Sports Certification

Activities & Achievements

- Leadership in college projects
- Participation in coding competitions
- Technical seminars and student clubs
- Sports and extracurricular activities

Step 8 - Verify Users

- ssh -i ~/.ssh/aws-key.pem ubuntu@16.16.122.140
- Then:
  cat /etc/passwd | grep user

```
ubuntu@ip-172-31-9-206:~$ cat /etc/passwd | grep user
user1:x:1001:1001::/home/user1:/bin/sh
user2:x:1002:1002::/home/user2:/bin/sh
ubuntu@ip-172-31-9-206:~$
```

```
ubuntu@ip-172-31-9-206:~$ ls /home
ubuntu  user1  user2
ubuntu@ip-172-31-9-206:~$
```

```
ubuntu@ip-172-31-9-206:~$ su user1
Password:
$ pwd
/home/ubuntu
$
```